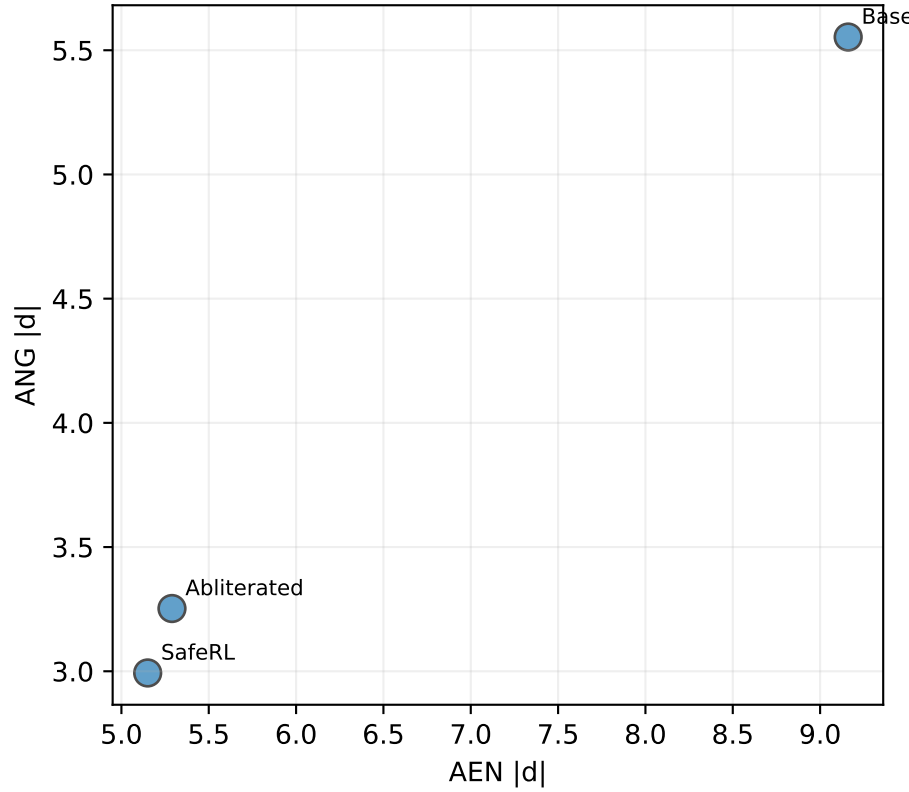
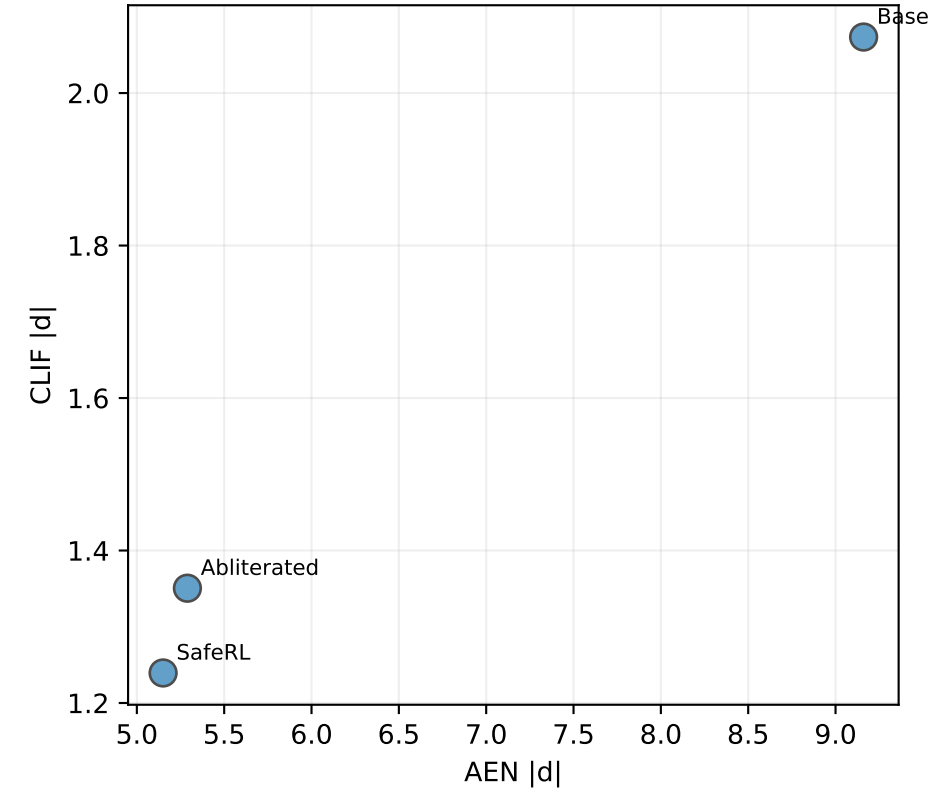


Cross-Metric Correlation: AEN vs Other Safety Metrics Qwen3-4B Lineage (Base, SafeRL, Abliterated)

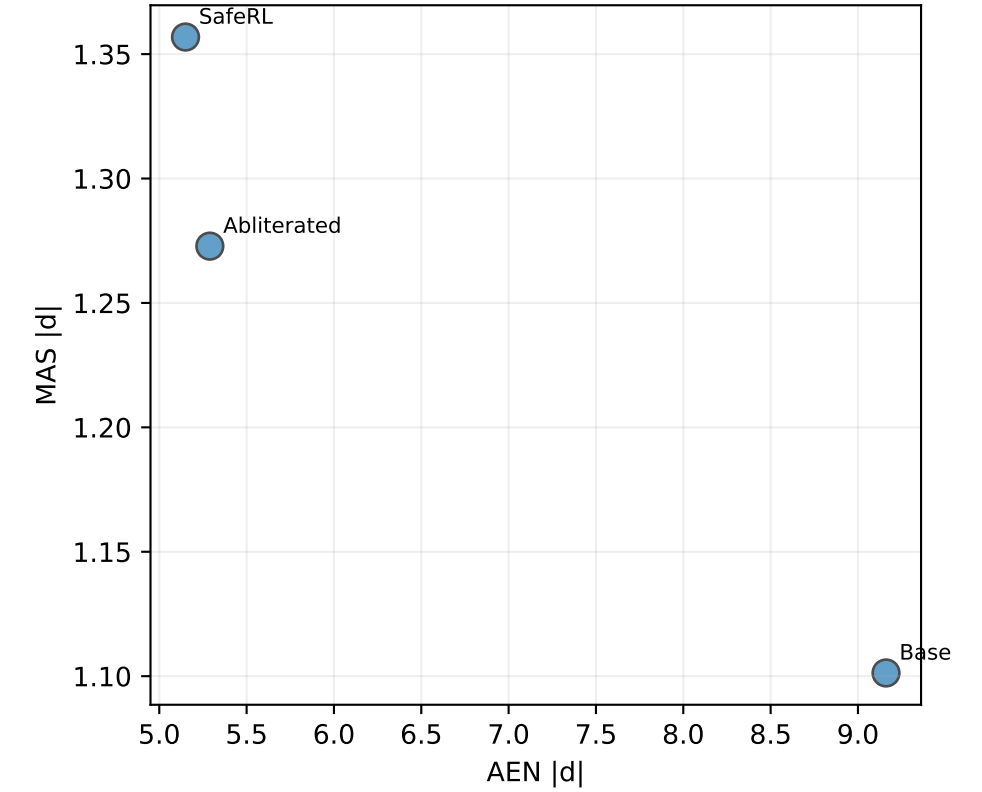
Activation Entropy (AEN) vs Activation Norm Gap (ANG)
 $r=0.998$, $p=0.039$



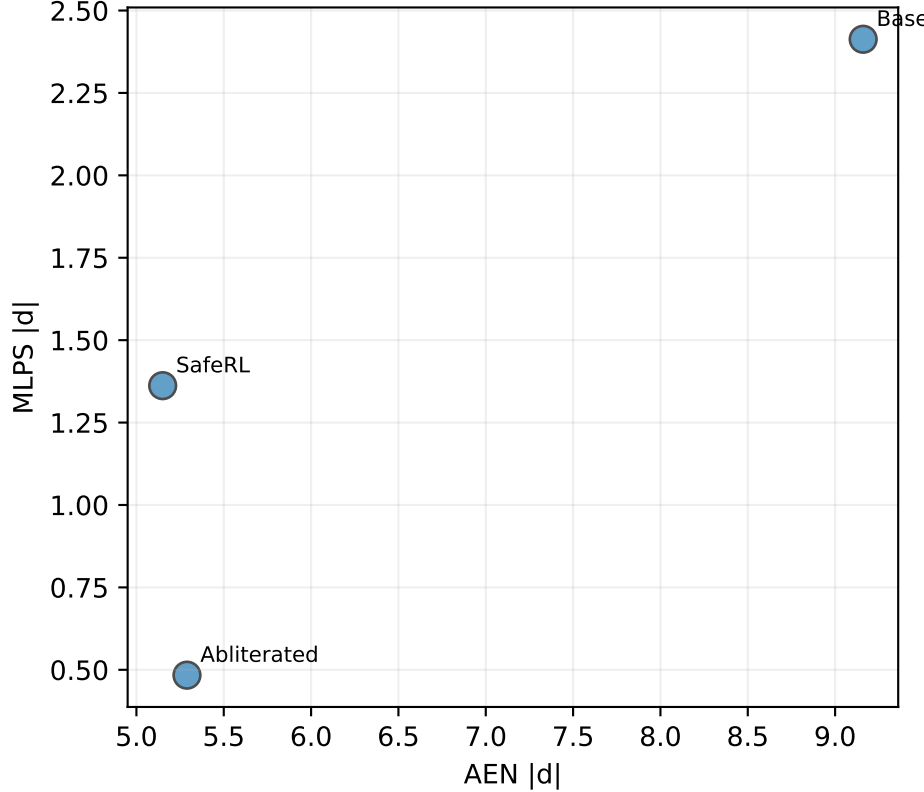
Activation Entropy (AEN) vs Cosine-Logit Invariant Feature (CLIF)
 $r=0.996$, $p=0.059$



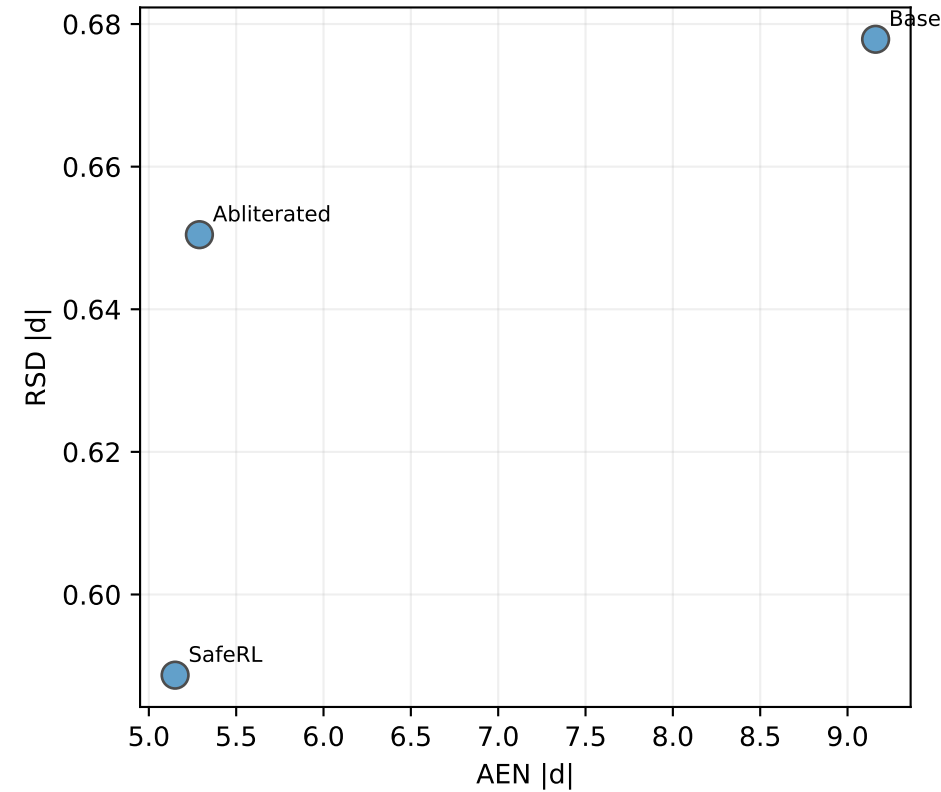
Activation Entropy (AEN) vs Mean Activation Similarity (MAS)
 $r=-0.956$, $p=0.190$



Activation Entropy (AEN) vs MLP Sparsity (MLPS)
 $r=0.876$, $p=0.320$



Activation Entropy (AEN) vs Residual Standard Deviation (RSD)
 $r=0.757$, $p=0.453$



Activation Entropy (AEN) vs Logit-Gap Baseline
 $r=-0.569$, $p=0.615$

